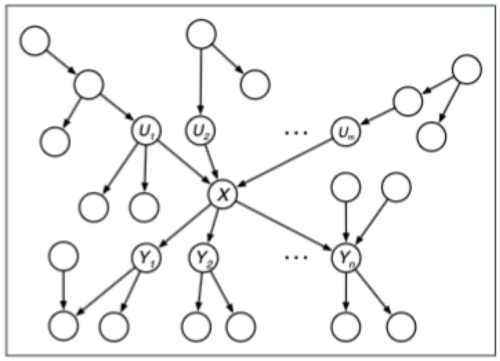
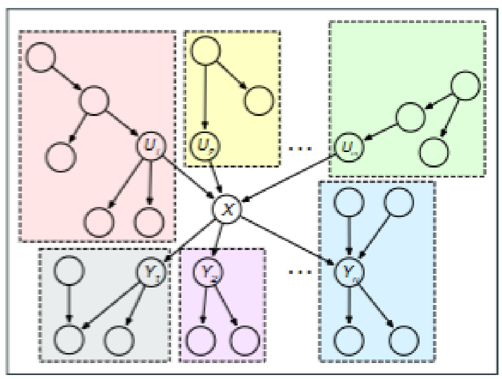
Polytree Exact Inference (Belief Propagation)

Bayesian networks

Suppose we have the following polytree shown below. Let X denote a node in the polytree with parents $U_1, U_2, ..., U_m$ and children $Y_1, Y_2, ..., Y_n$.



To perform the polytree algorithm, we first create the subgraphs around each parent U_i and child Y_j of X and all nodes connected to both not via X:



Now let E be the set containing all evidence nodes in the polytree. Suppose we want to compute

$$P(X = x|E). \tag{1}$$

Let E_X^+ and E_X^- denote the set of evidence nodes which are connected to X through the parents and children nodes respectively. Thus equation (1) becomes:

$$\begin{aligned}
P(X = x|E) &= P(X = x|E_X^+, E_X^-) && \boxed{\text{separating evidence}} \qquad (2)\\
&= \frac{P(E_X^-|X = x, E_X^+)P(X = x|E_X^+)}{P(E_X^-|E_X^+)} && \boxed{\text{Bayes rule}} \qquad (3)\\
&= \frac{P(E_X^-|X = x)P(X = x|E_X^+)}{P(E_X^-|E_X^+)} && \boxed{\text{conditional independence}} \qquad (4)
\end{aligned}$$

Lets break the fraction down and compute each term separately. We begin with $P(X = x|E_X^+)$. Let $\vec{U}$ denote the parents $(U_1, U_2, ..., U_m)$ of X and $\vec{u}$ denote values $(u_1, u_2, ..., u_m)$ for $\vec{U}$. Then $P(X = x|E_X^+)$ becomes:

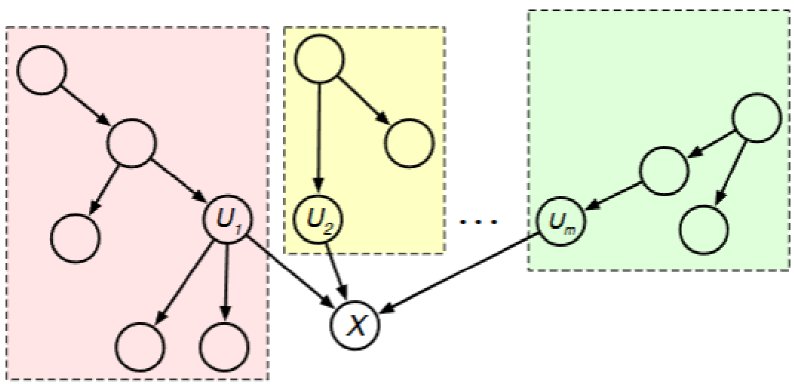
$$P(X = x|E_X^+) = \sum_{\vec{u}} P(X = x, \vec{U} = \vec{u}|E_X^+) \qquad \boxed{\text{marginalization}} \tag{5}$$

$$= \sum_{\vec{u}} P(\vec{U} = \vec{u}|E_X^+) P(X = x|\vec{U} = \vec{u}, E_X^+) \qquad \boxed{\text{product rule}} \tag{6}$$

$$= \sum_{\vec{u}} P(\vec{U} = \vec{u}|E_X^+) \underbrace{P(X = x|\vec{U} = \vec{u})}_{\text{CPT at node } X} \qquad \boxed{\text{conditional independence}} \tag{7}$$

Let $E_{U_i \backslash X} \subset E_X^+$ denote the evidence in the i^{th} parent's box; It is the evidence connected to U_i but not through X. Thus:

$$E_X^+ = E_{U_1 \backslash X} \cup E_{U_2 \backslash X} \cup ... \cup E_{U_m \backslash X} \tag{8}$$



Notice that if node X and its descendants are not observed, then X blocks every path between different boxes by case 3 of d-separation. Thus we can rewrite $P(\vec{U} = \vec{u}|E_X^+)$ in equation (7) as:

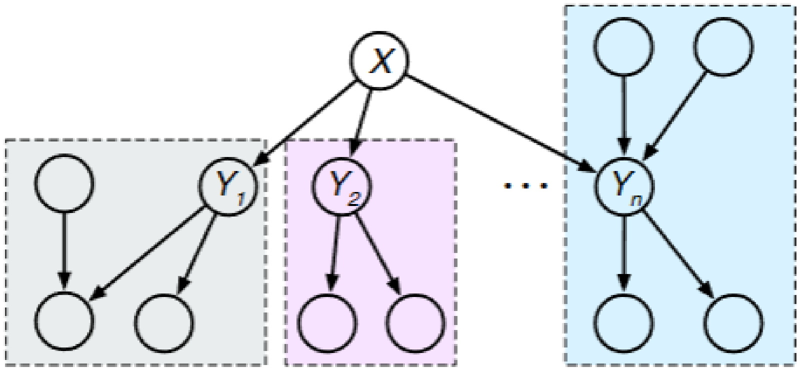
$$P(\vec{U} = \vec{u}|E_X^+) = \prod_{i=1}^{m} P(U_i = u_i|E_X^+) \qquad \boxed{\text{conditional independence}} \tag{9}$$

$$= \prod_{i=1}^{m} \underbrace{P(U_i = u_i|E_{U_i \backslash X})}_{\text{recursive instance}} \qquad \boxed{\text{conditional independence}} \tag{10}$$

Thus we can recurse up the polytree with equation (10) until we hit the root nodes. Putting it all together, $P(X = x|E_X^+)$:

$$P(X = x|E_X^+) = \sum_{\vec{u}} \underbrace{P(X = x|\vec{U} = \vec{u})}_{\text{CPT (given)}} \prod_{i=1}^{m} \underbrace{P(U_i = u_i|E_{U_i \backslash X})}_{\text{recursive instance}} \tag{11}$$

Next, we aim to simplify $P(E_X^-|X=x)$. To do so, we must recurse through the subgraphs of the children of X as seen in the figure below.

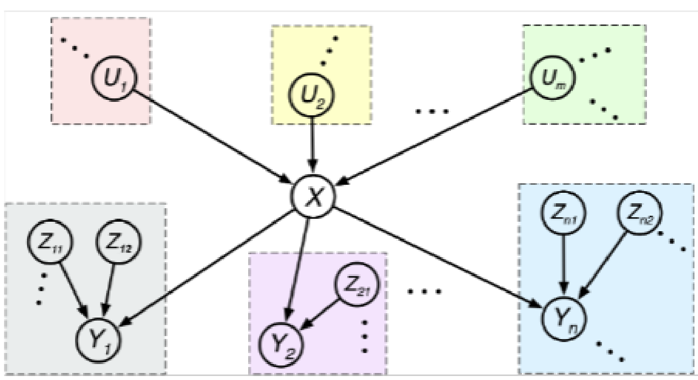


Notice that when X is observed, it blocks every path between the different subgraphs by case 2 of d-separation.

$$P(E_X^-|X=x) = \prod_{j=1}^{n} P(E_{Y_j \backslash X}|X=x) \quad \boxed{\text{conditional independence}} \tag{12}$$

Let Z_{jk} denote the k^{th} parent of Y_j (excluding X itself). Let $\vec{Z}_j$ denote the other parents of Y_j (excluding X). We can divide $E_{Y_j \backslash X}$ into two sets:

$$E_{Y_j \backslash X} = E_{Y_j}^- \cup E_{\vec{Z}_j \backslash Y_j} \tag{13}$$



$E_{Y_j}^-$ represents the set of all evidence nodes which are descendants of Y_j. $E_{\vec{Z}_j \backslash Y_j}$ denotes the set of all evidence nodes which are connected to Y_n through $\vec{Z}_j$. Thus equation (12) becomes:

$$
\begin{aligned}
P(E_X^-|X=x) &= \prod_{j=1}^{n} P(E_{Y_j}^-, E_{\vec{Z}_j\setminus Y_j}|X=x) && \boxed{\text{set separation}} \quad (14)\\
&= \prod_{j=1}^{n}\sum_{y_j}\sum_{\vec{z}_j} P(E_{Y_j}^-, E_{\vec{Z}_j\setminus Y_j}, Y_j=y_j, \vec{Z}_j=\vec{z}_j|X=x) && \boxed{\text{marginalization}} \quad (15)\\
&= \prod_{j=1}^{n}\sum_{y_j}\sum_{\vec{z}_j} P(E_{\vec{Z}_j\setminus Y_j}|X=x)P(\vec{Z}_j=\vec{z}_j|E_{\vec{Z}_j\setminus Y_j}, X=x)\\
&\quad P(Y_j=y_j|E_{\vec{Z}_j\setminus Y_j}, X=x, \vec{Z}_j) && \boxed{\text{product rule}} \quad (16)\\
&\quad P(E_{Y_j}^-|E_{\vec{Z}_j\setminus Y_j}, X=x, Y_j=y_j, \vec{Z}_j=\vec{z}_j)\\
&= \prod_{j=1}^{n}\sum_{y_j}\underbrace{P(E_{Y_j}^-|Y_j=y_j)}_{\text{recursive instance}}\sum_{\vec{z}_j}\underbrace{P(Y_j=y_j|X=x,\vec{Z}_j)}_{\text{CPT (given)}} && \boxed{\text{d-separation}} \quad (17)\\
&\quad \prod_{k}\underbrace{P(Z_{jk}=z_{jk}|E_{Z_{jk}\setminus Y_j})P(E_{Z_{jk}\setminus Y_j})}_{\text{recursive instance}}
\end{aligned}
$$

In equation (17), computing $P(E_{jk\setminus Y_j})$ will be extremely costly as it require us to recurse up and down the tree to find all evidence nodes in the set. As such, we can set $P(E_X^-|X=x)$ proportional to:

$$
P(E_X^-|X=x) \propto \prod_{j=1}^{n}\sum_{y_j}\underbrace{P(E_{Y_j}^-|Y_j=y_j)}_{\text{recursive instance}}\sum_{\vec{z}_j}\underbrace{P(Y_j=y_j|X=x,\vec{Z}_j)}_{\text{CPT (given)}}\prod_{k}\underbrace{P(Z_{jk}=z_{jk}|E_{Z_{jk}\setminus Y_j})}_{\text{recursive instance}} \quad (18)
$$

Looking back at equation (4), we just need to compute the denominator term $P(E_X^-|E_X^+)$. However, this is also extremely costly as it will force us to run the polytree multiple times for each specific outcome of X. Since this normalizing term does not change the relationship of the numerator term, we can set $P(X=x|E)$ proportional to:

$$
P(X=x|E) \propto P(E_X^-|X=x)P(X=x|E_X^+), \quad (19)
$$

Where $P(E_X^-|X=x)$ is proportional to equation (18) and $P(X=x|E_X^+)$ is equal to equation (11).